Algorithms for DNA Sequencing

Today's goal • 3/3 items

Module 1
DNA sequencing, strings and matching

Welcome

Welcome to Algorithms for DNA Sequencing
Reading • 10 min

Pre Course Survey
Reading • 10 min

Syllabus
Reading • 10 min

Setting up Python (and Jupyter)
Reading • 10 min

Getting slides and notebooks
Reading • 10 min

Using data files with Python programs
Reading • 10 min

Module 1: DNA sequencing, strings and matching

Module 1 Introduction
Video • 1 min

Lecture: Why study this?
Video • 4 min

Lecture: DNA sequencing past and present
Video • 3 min

Lecture: Genomes as strings, reads as substrings
Video • 5 min

Lecture: String definitions and Python examples
Video • 3 min

Practical: String basics
Video • 7 min

Practical: Manipulating DNA strings
Video • 7 min

Practical: Downloading and parsing a genome
Video • 6 min

Lecture: How DNA gets copied
Video • 3 min

Optional lecture: How second-generation sequencers work
Video • 7 min

Optional lecture: Sequencing errors and base qualities
Video • 6 min

Lecture: Sequencing reads in FASTQ format
Video • 4 min

Practical: Working with sequencing reads
Video • 11 min

Practical: Analyzing reads by position
Video • 6 min

Lecture: Sequencers give pieces to genomic puzzles
Video • 5 min

Lecture: Read alignment and why it's hard
Video • 3 min

Lecture: Naive exact matching
Video • 10 min

Practical: Matching artificial reads
Video • 6 min

Practical: Matching real reads
Video • 7 min

Quiz

Module 1
Graded Assignment • 30 min

Programming Homework

Programming Homework 1 Instructions (Read First)
Reading • 10 min

Programming Homework 1
Graded Assignment • 30 min

Module 2
Preprocessing, indexing and approximate matching

Module 3
Edit distance, assembly, overlaps

Module 4
Algorithms for assembly

Using data files with Python programs

0:01 / 1:10 2x

In practical sessions and programming assignments, you will sometimes work with data files we provide. These might be reference genomes in FASTA format (e.g. lambda_virus.fa), sequencing reads in FASTQ format (e.g. SRR835775_1.first1000.fastq), or Python modules (e.g. kmer_index.py). To use a file with your Python code, you have to (a) download the file to the same computer as your Python program; use a web browser or a command-line tool like wget or curl to download it, then (b) put the file in the same directory with your Python program. Then you can open it from your Python program using the Python open function.

If you use Jupyter, another way to do this (which we use in the practical sessions) is with a command that starts with !wget. This is a shell command. But if you are using try.jupyter.org note that the !wget command will not work ("unable to resolve host address"). In this case, we recommend you either (a) use www.wakari.io instead, or (b) use the "upload" button in the try.jupyter.org interface to upload the file to the server, then skip the command that begins with !wget.

Best,

Ben

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →